

EXPERIMENT 6

DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize the City Day Air Quality dataset using Power BI Desktop and create interactive dashboards and reports for decision-making.

Algorithm 1

Importing Data into Power BI

Step 1 Open Power BI Desktop

Step 2 Select: Home → Get Data

Step 3 Choose Text/CSV

Step 4 Browse the **city_day.csv** dataset and select it

Dataset: City Day Air Quality Dataset **Rows:** 29,531 **Columns:** 16

Algorithm 2

Data Transformation using Power Query

Step 1 Select Transform Data

Step 2 Open Power Query Editor

Step 3 Remove duplicate records

Step 4 Handle missing/null values

Step 5 Rename or standardize columns where required

Step 6 Apply changes

Procedure

1. The data preview window will appear. Select **Transform Data** instead of Load to open the Power Query Editor.
2. Examine the imported table and verify the required columns: City, Date, PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene, AQI and AQI_Bucket.
3. Remove completely blank rows using Home → Remove Rows → Remove Blank Rows.

4. Select all columns and choose Home → Remove Rows → Remove Duplicates. The uploaded dataset contains 29,531 records after duplicate removal.

5. Standardize the **City** column using Transform → Format → Capitalize Each Word, so city names use a consistent representation.

6. Select **Date** and choose Transform → Data Type → Date. This converts the date field from text/object format to a proper date.

7. Inspect the **AQI_Bucket** column and retain the provided categories. Missing category values are left identifiable rather than inventing a category.

Dataset date range: 01-01-2015 to 01-07-2020

Number of cities: 26

8. Handle missing numeric values according to the analysis requirement. For Power BI visuals, exclude blank measurements from averages rather than assigning an artificial value.
9. Check the AQI column for missing observations. The dataset contains **4,681** missing AQI values.
10. Verify that pollutant columns are numeric: PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene and Xylene.
11. Verify the AQI_Bucket field as Text and City as Text.
12. Review the Applied Steps pane and correct any transformation errors before applying the query.

13. After completing the cleaning and transformation operations, select **Home** → **Close & Apply**. Power BI applies the transformations and loads the cleaned table into the data model.

14. Switch to Data/Table view and inspect the transformed data. Verify that duplicate rows, blank rows, inconsistent city names, invalid dates and missing values have been handled correctly.

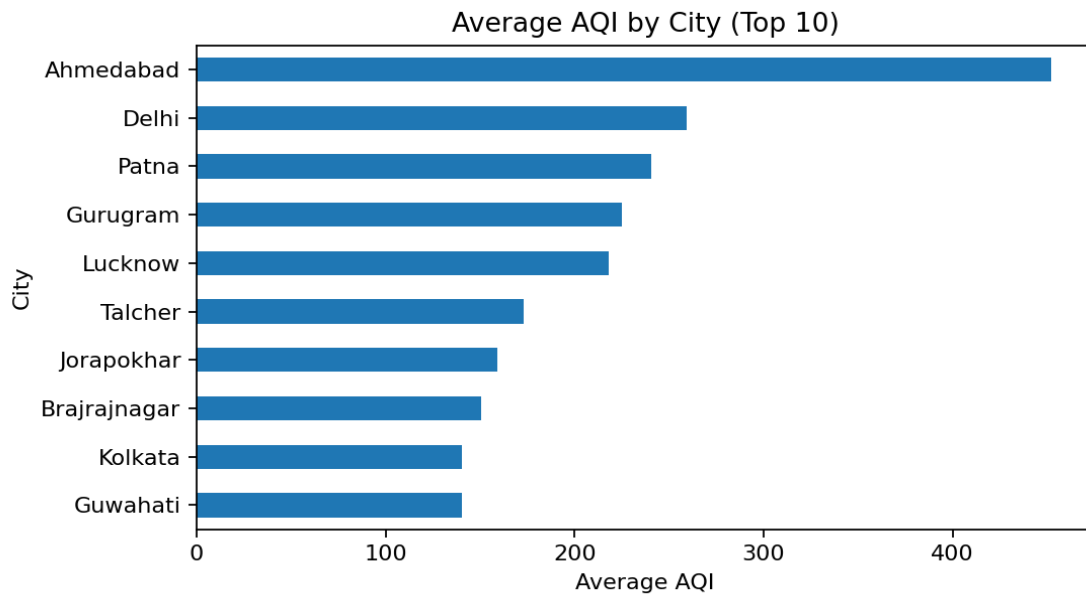
Recommended data types

Column	Data Type
City	Text
Date	Date
Pollutant columns	Decimal Number
AQI	Decimal Number
AQI_Bucket	Text

15. To verify the ETL process visually, switch to Report view and create a **Clustered Bar Chart**. Drag **City** to the category axis and **AQI** to Values, using Average as the aggregation.

16. Sort the visual in descending order of Average AQI. This provides a city-level comparison of air quality.

Output:



The resulting visual confirms that the cleaned City Day data can be loaded into Power BI and used for comparative air-quality analysis.

Interpretation:

Ahmedabad has the highest average AQI among the cities in the dataset (452.12).

Aizawl has the lowest average AQI among the cities with available AQI values (34.77).

Algorithm 3

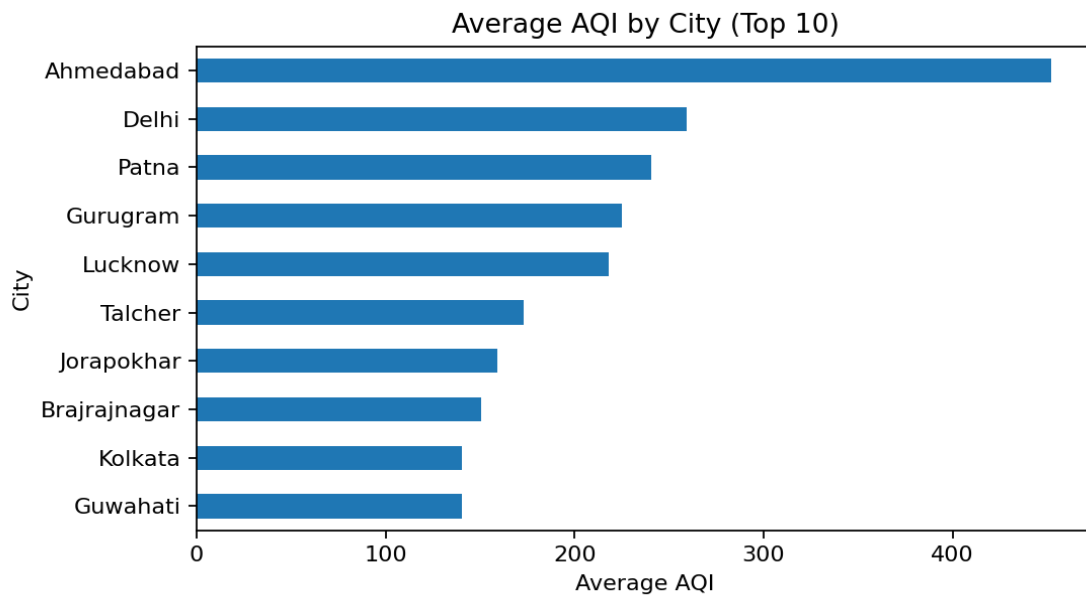
Creating a Bar Chart

Step 1 Choose Bar Chart Visualization

Step 2 Drag **City** → Y-Axis

Step 3 Drag **AQI** → X-Axis and set aggregation to Average

Step 4 Format chart and sort by Average AQI



Algorithm 4

Creating a Pie Chart

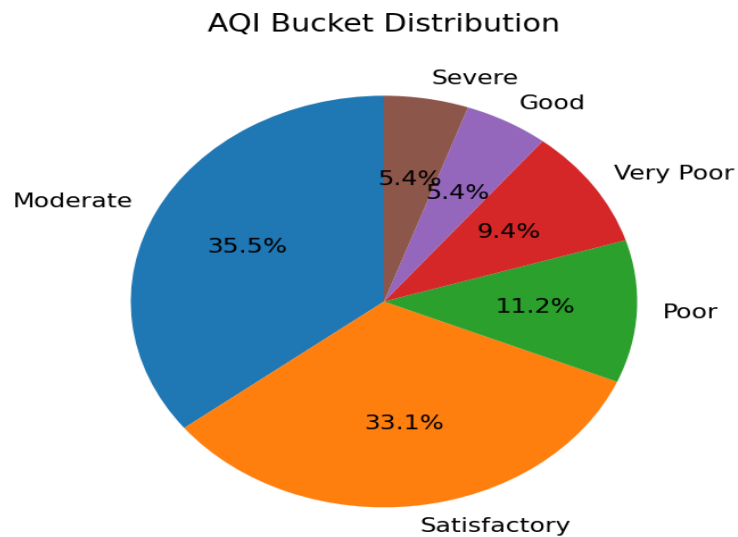
Step 1 Select Pie Chart

Step 2 Drag **AQI_Bucket** → Legend

Step 3 Drag **AQI_Bucket** → Values and use Count

Step 4 Format data labels to show percentage

Output:

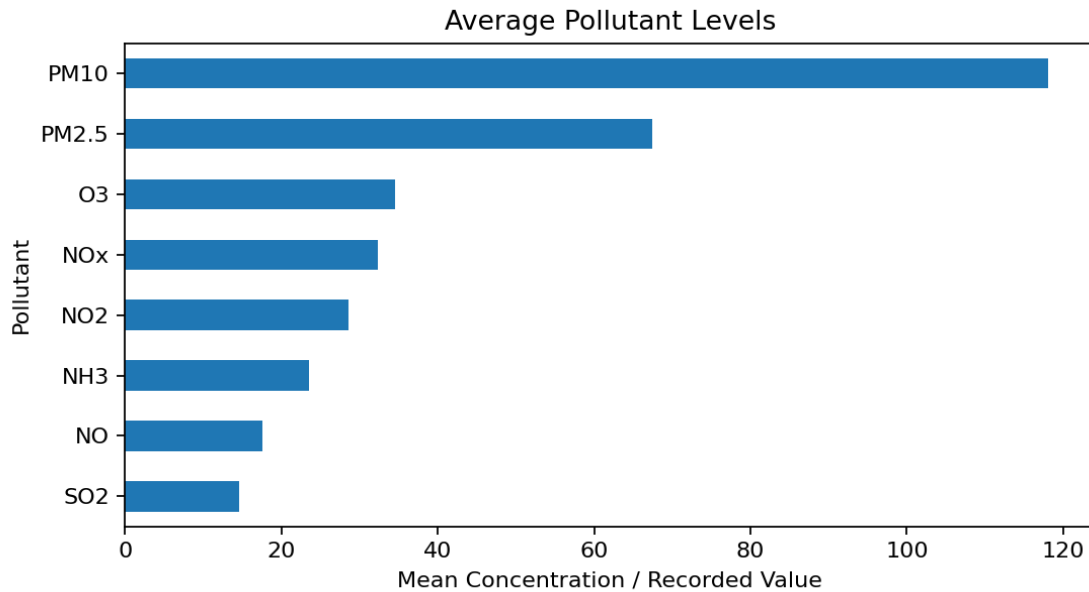


Interpretation:

The **Moderate** AQI category has the largest number of recorded observations, representing approximately **35.5%** of non-missing AQI bucket records.

Additional pollutant analysis

Create a Bar Chart with pollutant fields as categories and their Average values to compare the recorded pollutant levels.



Interpretation:

Among the listed pollutant variables, **PM10** has the highest mean recorded value in the uploaded dataset.

A line chart can also be created using Date on the X-axis and Average AQI on the Y-axis to observe changes over time.

Algorithm 5

Creating a KPI Card

Step 1 Create measures by clicking **New Measure** from the Measure tools tab.

DAX measures

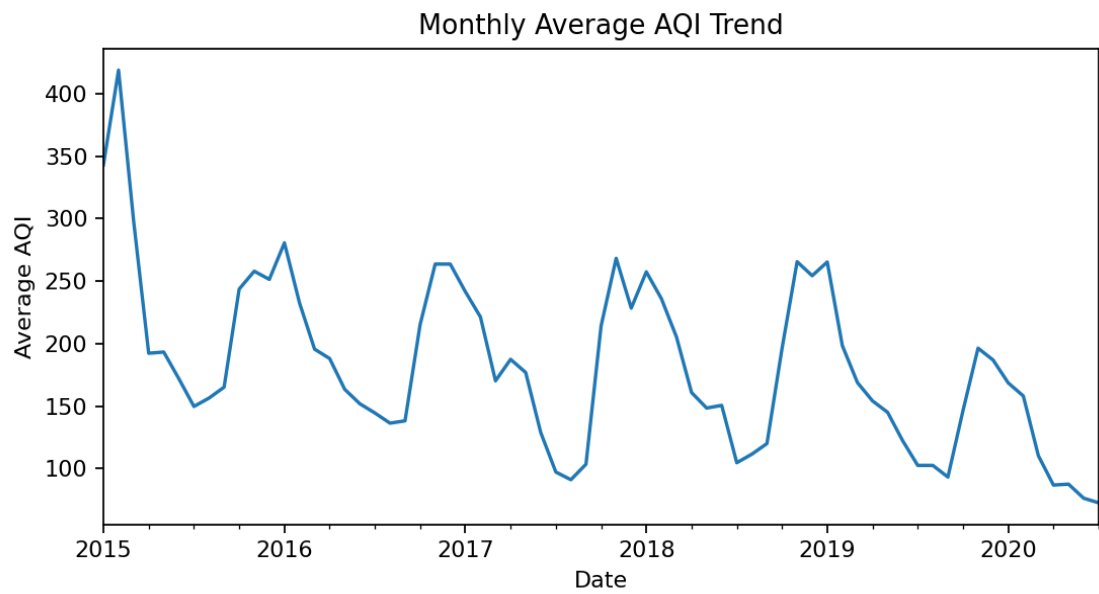
Measure	DAX Formula
TotalRecords	COUNTROWS('city_day')
AverageAQI	AVERAGE('city_day'[AQI])
MaximumAQI	MAX('city_day'[AQI])
CityCount	DISTINCTCOUNT('city_day'[City])

Step 2 Create Card visuals for each measure

Step 3 Drag the corresponding measure into the field well

Output:

KPI	Dataset Value
Total Records	29,531
Average AQI	166.46
Maximum AQI	2049
Number of Cities	26



Interpretation: The KPI cards summarize the size of the dataset and the overall AQI statistics, while the line chart provides a time-based view of air-quality variation.

Algorithm 6

Creating Slicers

Step 1 Select Slicer Visualization

Step 2 Drag **City** field into the slicer

Step 3 Enable Single Selection when a single-city analysis is required

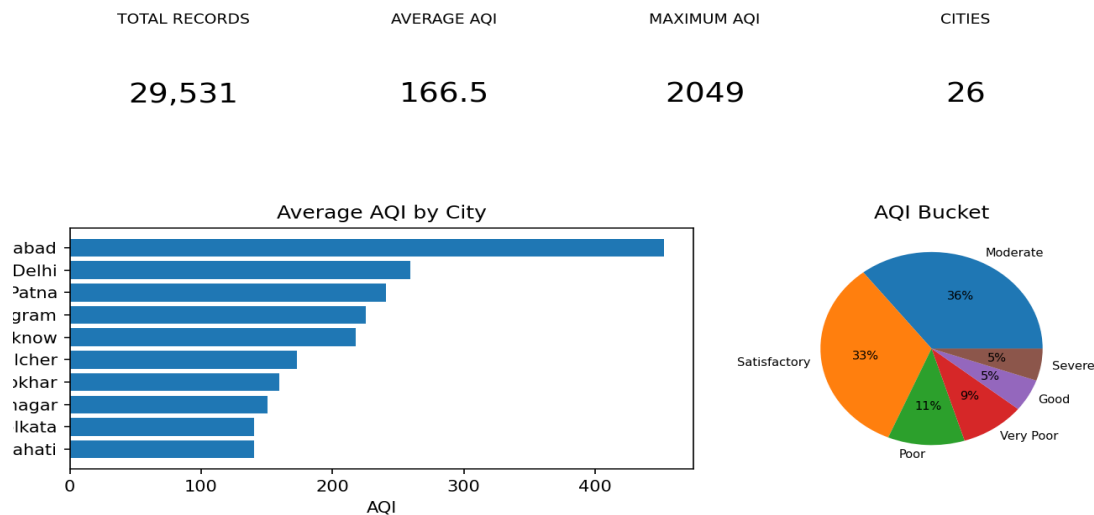
Step 4 Apply filters

Output:

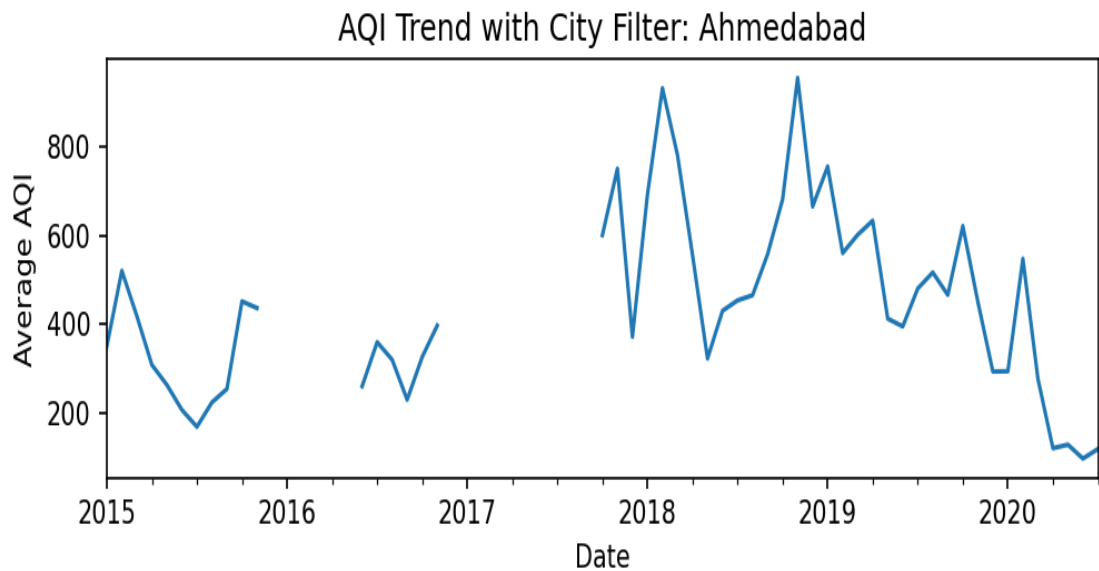
CITY SLICER
Ahmedabad
Delhi
Patna
Gurugram
Lucknow
...

Selecting a city in the slicer filters the KPI cards and charts on the report page.

Without any filter:



With a city filter:



When the City slicer is applied, the report updates to show only the selected city's records and measures.

Algorithm 8

Creating Dashboard

Step 1 Add KPI Cards

Step 2 Add City-wise AQI Bar Chart

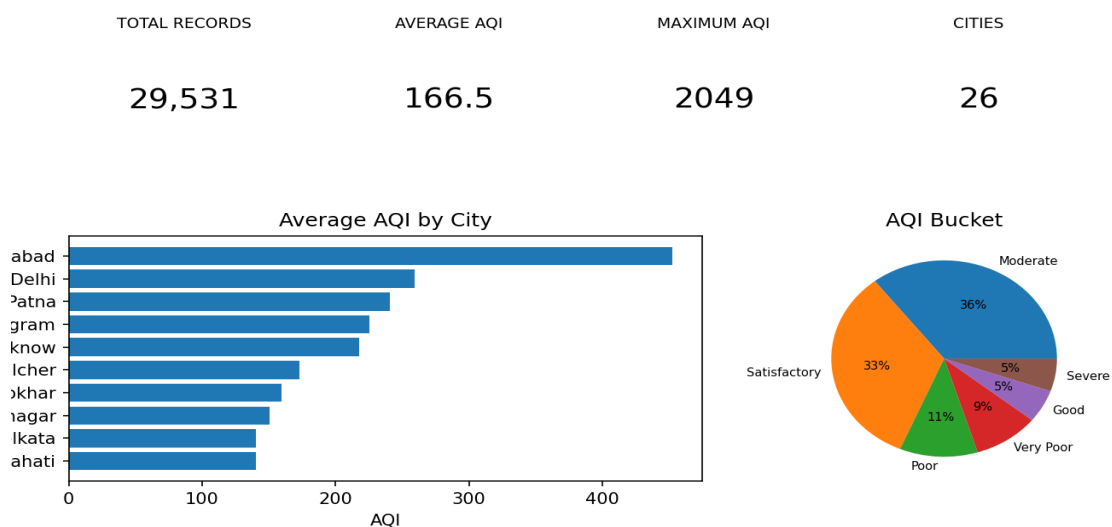
Step 3 Add AQI Bucket Pie Chart

Step 4 Add City Slicer

Step 5 Arrange and format visuals

Step 6 Save the Power BI report/dashboard

Output:



Result:

The City Day air-quality data was successfully prepared for import, transformation, analysis, and visualization in Power BI Desktop. Interactive KPI cards, city-wise AQI comparison, AQI category distribution, trend analysis, and a City slicer were created to support effective decision-making.